

Assignment: Real-Time 3D Scene Reconstruction with 3D Gaussian Splatting

1. Overview

In this assignment, you will perform a complete end-to-end 3D reconstruction pipeline using **3D Gaussian Splatting (3DGS)**. You will process a provided image sequence, use **COLMAP** to estimate camera poses, train a 3DGS model, and evaluate its performance on a Novel View Synthesis (NVS) task.

2. Learning Objectives

- Configure the 3DGS environment using either Conda or Docker.
- Understand Structure-from-Motion (SfM) through COLMAP for camera pose estimation.¹
- Implement the 3DGS training and optimization pipeline.³
- Evaluate reconstruction quality using standard metrics: PSNR, SSIM, and LPIPS.⁴
- Visualize explicit radiance fields using real-time viewers.³

3. Timeline & Submission

- **Deadline:** You have **two weeks** from the date of assignment to complete the project.
- **Submission:** All deliverables must be submitted via **Canvas**.
- **Deliverable:** A **1-page Google Doc report (written in English)** summarizing your results and workflow.

4. Project Requirements

- **Dataset:** You must use the prepared image datasets provided on Canvas: **files/Homeworks/HW1**. Do **not** use custom data.³
- **Evaluation Split:** You must use the "1-in-8" test split (every 8th image is reserved for testing, with the remaining images used for training).⁴

5. Implementation Steps

Task 1: Environment Setup

You can set up the 3DGS environment using one of the following two methods. Both require an NVIDIA GPU with CUDA support (version 11.6 or higher recommended).

- **Option A: Conda**
 1. Clone the repository recursively: `git clone --recursive <repo_url>`.
 2. Create the environment: `conda env create -f environment.yml`.³
 3. Activate: `conda activate gsplat`.³
- **Option B: Docker**
 1. Use a compatible 3DGS Docker image (e.g., `eduardtalberg/gaussian_splatting`).
 2. Ensure the NVIDIA Container Toolkit is installed on your host machine.
 3. Run the container with GPU access enabled:
`docker run --gpus all -v <local_path>:/workspace/data <image_name>`.

Task 2: Camera Pose Estimation (COLMAP)

Use COLMAP to generate the initial sparse point cloud and camera parameters required for Gaussian initialization.¹

- **Preprocessing:** Run the `convert.py` script to automate feature extraction, matching, and image undistortion: `python convert.py -s <path_to_HW1_data>`.³
- **Verification:** Ensure the `sparse/0` directory is generated, containing the necessary `cameras.bin`, `images.bin`, and `points3D.bin` files.¹

Task 3: Training the 3DGS Model

Execute the training script. Using the `--eval` flag will automatically handle the "1-in-8" data split for evaluation.⁴

- **Command:** `python train.py -s <path_to_HW1_data> --eval`.⁴
- **Iterations:** Training typically defaults to 30,000 iterations, though you may observe convergence as early as 7,000 iterations for small datasets.

Task 4: Evaluation and Visualization

1. **Quantitative Metrics:** Render the test views and compute the PSNR, SSIM, and LPIPS metrics: `python render.py -m <path_to_model> --eval`.³ `python metrics.py -m <path_to_model>`.⁴
2. **Visualization:** Use a real-time viewer (e.g., SIBR or browser-based SuperSplat) to observe the 3D scene.³ Capture a "Novel View" (from the test set) and a "Global Rendering" for your report.



6. Report Template (1-Page English Report)

Your report must be a single-page document in **English** covering:

1. **Environment:** State whether you used Conda or Docker and list your GPU model.⁸
2. **Data Split:** Confirm the number of training vs. testing images.⁴
3. **Metrics Table:**

Metric	Value	Reference Target
PSNR		> dB ¹⁰
SSIM		> 12
LPIPS		Lower is better ¹³

4. **Screenshots:**
 - **Global View:** A shot showing the overall reconstructed scene.
 - **NVS Comparison:** Place the Ground Truth test image next to your 3DGS render of that same viewpoint.¹⁴
5. **Brief Analysis:** 2-3 sentences discussing any artifacts (e.g., "floaters" or blur) observed and the efficiency of the training process.¹⁵

References

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15. 3D Gaussian Splatting: A Technical Guide to Real-Time Neural Rendering - KIRI Engine, accessed February 4, 2026, <https://www.kiriengine.app/blog/3d-gaussian-splatting-a-technical-guide-to-real-time-neural-rendering>